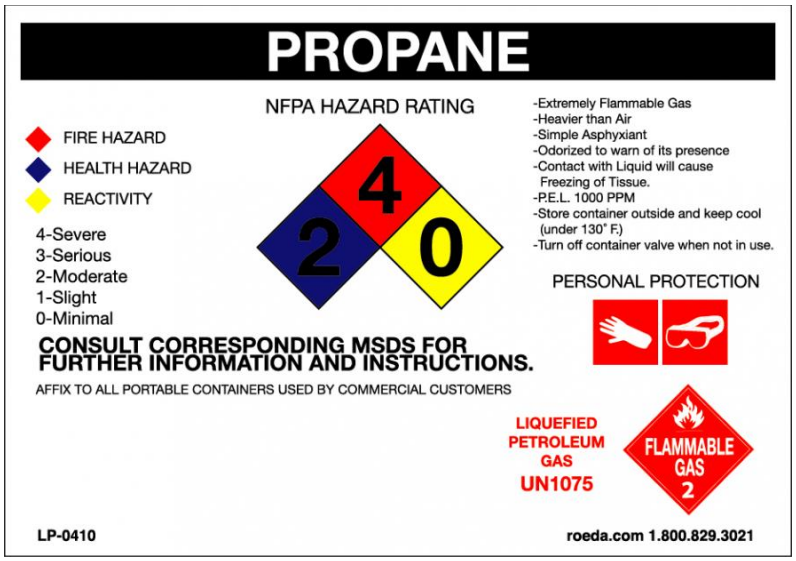
Chemical Safety (Propane)

Properties:



- Propane's formula is C_3H_8. It is an alkane.
- Propane is a gas at Standard temperature pressure but can be compressed to a transportable liquid (in this state it is not flammable)
- The density of propane is 1.898Kg/m^3, which is greater than air
- The Odor threshold of propane is 20000 ppm
- The OSHWA PEL exposure limit for propane is 1000 ppm
- The flashpoint of propane is -156°F / -104°C
- The molecular weight of propane is 44.1 g/mol
- Boiling point of Propane is -46.6^{o}F / -42°C
- Propane is non-Toxic but highly Flammable
- Ignition temperature of Propane 940°F / 504.44^{o}C

How is propane supposed to be stored?

Propane should always be stored outdoors, in a flat level area out, upright and out of direct sunlight. It should also be kept 10 feet away from flammable materials. This is so that if a valve is loose and some propane escapes, propane (which is 1.55 times heavier than air), does not concentrate indoors as it is highly susceptible to ignition. The temperature for storage should be above -40°C/F and below 120°F / 49°C. Propane is also an asphyxiant which displaces oxygen in the lungs and suffocates victims who have inhaled high concentrations of propane.

What should propane be stored in?

Propane is typically stored in a welded steel tank; this is because steel is a very strong material and is easily available. It also is strong enough to withstand the vapor pressure of propane which is usually transported in a liquid state. Propane is transported in a liquid form in which it is not flammable.

$$C_3H_8(s) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

Above is the combustion reaction of propane

Risk safety analysis:

Scenario	Risk	Level of consequences	Safety Precautions	Risk of failure	Backups
The valve of a propane tank is left open connected to a Bar-B-Que Pit.	Someone can ignite exposed gas by passing with an open flame.	If this happens, this can cause an explosion and damage or kill anyone in the vicinity.	The bar-b-que Pitt has valves of its own that stops gas from flowing unless it is on	Medium as these valves may not be designed specifically to stop the flow of propane for extended periods of time as they assume the propane tank have valves themselves	The BBQ pit is stored outdoors so propane would not be allowed to concentrate
Propane tank is stored next to a fireplace.	The tank ignites due to heat from the fireplace.	If this happens, this can cause an explosion and damage or kill anyone in the vicinity.	Valve stops gas from leaving tank.	Medium, because while valve is built strongly, there is a chance that someone opens the valve for various reasons	There are none.
Tank of propane falls from the back of a truck	When the tank falls, a spark is caused, and tank ignites	This can be high as if this happens on the road, this can cause an accident killing drivers, If the other tanks are ignited by the explosion, it can kill many in the vicinity	Propane is transported as a liquid, a state in which propane is nonflammable	This is high, as propane tanks are not filled to the brim to allow pressure to equalize in the tank and prevent the tank from bursting via vapor pressure. This means there is some gas in the tank which can ignite	The tank is made of steel, which is a very strong material that is very unlikely to fail and expose gas.